

TA Analyzer - Detailed Guide

This document is the detailed manual scoped to the five slides of the attached `Manual.pptx`. For each topic it covers **purpose** · **prerequisites** · **step-by-step** · **options** · **checks** · **common warnings**.

Slide	Topic	Section
5	Program overview / main window	§0
1	Load Data (Custom) - manual selection	§1
2	Background Correction	§2
3	Chirp Correction	§3
4	Kinetic Fit (single trace)	§4

§0. Program overview (Slide 5)

0.1 Launching the program

- 1 Start **Anaconda Prompt** (Windows Start menu → "Anaconda Prompt").
- 2 Use `cd` to navigate to this project's folder.
 - Example: `cd C:\Users\watqd\OneDrive\KNU_ULSIL\Python_Code\TA_Analyzer_rev5`
- 3 Run `python ta_main.py` to launch the program.

Launching from **Anaconda Prompt** (rather than a plain python shell) ensures that numpy / scipy / PyQt5 / matplotlib and the other dependencies load correctly.

0.2 Main window layout

After launch the main window has three primary panels.

- **Large left panel** - 2D ΔA map (x: wavelength, y: delay time, color: ΔA)
- **Top right - Spectrum** (ΔA vs wavelength) at the selected delay
- **Bottom right - Kinetics** (ΔA vs delay) at the selected wavelength

The top of the window has two toolbar rows. The buttons used in this manual are:

Button	Purpose
Load Data...	Load a CSV/DAT/TXT in a recognised standard format (auto-detect)
Load Data (Custom)...	Manually pick X / Y / Z regions for non-standard CSV layouts
Background Correction...	Subtract the mean of the first N delay points as background
Chirp Correction...	Click t points → fit → apply chirp correction
More Analysis	Submenu with SVD, Kinetic Fit , LDA, MCR-ALS, Coherence

0.3 Data-loading flow

- 1 Click **Load Data...** - if the format is recognised, the file loads immediately.

- 2 If detection fails or the layout is unusual, use **Load Data (Custom)...** (see §1).
- 3 Once loaded, the 2D map and the side spectrum / kinetics panels refresh automatically.

Selecting a single file loads it directly; selecting several files opens the **Load & Average** window.

§1. Load Data (Custom) - manual X / Y / Z region selection (Slide 1)

1.1 Purpose

For measurement files whose header layout the automatic parser cannot recognise, this dialog presents the CSV as an **Excel-like spreadsheet preview** so you can drag-select the locations of wavelength (X), delay (Y), and ΔA (Z) by hand.

1.2 Prerequisites

- File format: **CSV (comma / tab / semicolon delimited; whitespace also recognised)**
- X (wavelength) and Y (delay) must be stored as **1D arrays - a single row or single column**.
- The Z matrix must have shape `len(X) × len(Y)` or its transpose `len(Y) × len(X)`.

1.3 Dialog layout

From top to bottom:

- 1 **File path / Browse...** - replace the previewed file.
- 2 **Stage banner (blue)** - current step indicator (**STEP 1 of 3 ...**).
- 3 **Current selection** - live readout of what the mouse has selected (**A1:K1, n rows × m cols**).
- 4 **Table (Excel-style)** - cell color
 - white: numeric cell
 - yellow: non-numeric (header / metadata)
 - gray: blank cell
- 5 **Shortcut hint line** below the table.
- 6 Buttons: **Confirm This Selection / Clear Current Selection / Back to Previous Step / Reset All / Auto-detect**.
- 7 **Confirmed selections** panel - addresses and sizes of the X / Y / Z ranges fixed so far.
- 8 **Load / Cancel** - **Load** is enabled only after all three regions are confirmed.

1.4 Procedure

The order is fixed at **X → Y → Z**, and each step must pass validation before the dialog advances.

STEP 1: select the X-axis (wavelength)

- Drag-select **one row OR one column** containing the wavelengths.
- Click **Confirm This Selection**.
- Validation
- 2D selections (more than one row AND more than one column) are rejected → "X-axis must be a 1D array."
- Selections with blank or non-numeric cells are rejected → "Non-numeric / Too Short".

STEP 2: select the Y-axis (delay)

- Drag-select a single row or single column that holds the delays → **Confirm This Selection**.

- Same validation criteria as X (1D, every cell is a finite number).

STEP 3: select the Z-matrix (ΔA)

- Drag-select the rectangular 2D block of ΔA values → **Confirm This Selection**.
- Validation
- If the shape is neither $(\text{len}(X) \times \text{len}(Y))$ nor $(\text{len}(Y) \times \text{len}(X))$, the dialog rejects it → "Z-matrix shape ($n \times m$) does not match the X-axis (X points) and Y-axis (Y points)."
- You may select the transposed orientation - on **Load** the matrix is automatically rotated to $(\text{wavelength} \times \text{delay})$.

Finish

- When all three regions are confirmed, the stage banner reads "All three regions confirmed" and the **Load** button enables.
- Click **Load** to push the data into the main window.

1.5 Helper actions

Action	Description
Clear Current Selection	Clear only the table highlight at the current step (confirmed steps stay)
Back to Previous Step	Go back one step and discard that step's confirmed value
Reset All	Clear X/Y/Z and return to STEP 1
Auto-detect	Fill X/Y/Z with the built-in auto parser (correct manually if wrong, then Load)

1.6 Keyboard shortcuts (Excel-compatible)

The following keys work while focus is on the preview table.

Key	Action
Drag	Standard region selection
Shift + Click	Extend the selection from the anchor to the clicked cell
Ctrl + ↑ / ↓ / ← / →	Jump the cursor to the data edge (or the next data block)
Ctrl + Shift + ↑ / ↓ / ← / →	Extend the selection from the anchor to the data edge

Ctrl + Shift + Arrow rules (Excel-equivalent) - Current cell is data and the next neighbor is data: extend to the last cell before a blank. - Current cell is data and the next neighbor is blank: extend to the next data block. - Current cell is blank: extend to the next data cell. - If no further data exists in that direction, extend to the table boundary.

1.7 Common warnings

Message	Cause / fix
X/Y-axis must be a 1D array.	Both row and column lengths are greater than 1. Select only one row or one column .
Non-numeric / Too Short	The selection contains blank cells or text. Exclude header cells and try again.
Z-matrix shape ($n \times m$) does not match the X-axis ... and Y-axis ...	Z's row / column counts differ from X and Y. Use Reset All and start from STEP 1.

§2. Background Correction (Slide 2)

2.1 Purpose

Estimate the pre-time-zero ($t < 0$) average spectrum as a background and subtract it from every delay.

2.2 Prerequisites

- A dataset must be loaded ($N \lambda \times N t$ should appear in the status line of the main window).

2.3 Dialog layout

- **# of initial delay points to average** - number of initial delay points to average (spinner or direct input).
- Side label: **Using $t = \dots \dots$ ps** - informs which actual time interval the chosen count corresponds to.
- Center graph: ΔA spectra for the first N delays (thin colored lines) plus the average (thick black line). A flat average is desirable.
- Bottom: **Apply & Close**, **Cancel**.

2.4 Procedure

- 1 Click **Background Correction...** on the toolbar.
- 2 Adjust the **# of initial delay points to average** value.
 - Too small $\rightarrow$ noise gets baked into the background; too large $\rightarrow$ real near-time-zero signal contaminates it.
 - Aim for a range where **Using $t = a \dots b$ ps** stays entirely in $t < 0$ (before the pump arrives).
- 3 Verify that the average (black line) is relatively flat around zero.
- 4 Click **Apply & Close** - this average is subtracted from the entire ΔA matrix.

2.5 Undo / re-apply

- If applied incorrectly, **Reset Corrections** on the main toolbar reverts background (and every other) correction at once.
- To re-apply with a different value, reopen **Background Correction...** and click **Apply & Close** again.

§3. Chirp Correction (Slide 3)

3.1 Purpose

In femtosecond pump-probe measurements, group-velocity dispersion of the white-light continuum makes $t = 0$ wavelength-dependent. The user clicks the true t at multiple wavelengths; the dialog fits those points and corrects every wavelength's delay axis accordingly.

3.2 Prerequisites

- A dataset must be loaded.
- The coherent artifact / IRF feature near $t = 0$ should be visible so the clicks can be placed accurately.

3.3 Dialog layout

The 2D map (for picking) is on the left and the fit-result table / residual graph are on the right.

Region	Description
View t_{min} / t_{max}	Delay range shown on the map. Narrow to the chirp region (e.g. 0.5 ~ 1.5 ps) for accurate clicks.
Auto-place	Place t points automatically via ridge detection
Recompute	Recompute when ridge-detection parameters change
Reset view	Restore the time range and the fit-result graph to defaults
Remove Last / Clear Points	Remove the most recent / all picked points
Fit & Apply	Fit the current points and apply chirp correction to the data

3.4 Procedure

- 1 Click **Chirp Correction...** on the toolbar.
- 2 Adjust **View t_{min}** and **View t_{max}** to focus on the chirp.
- 3 On the 2D map, click each wavelength's t (the leading edge of the coherent artifact) in turn.
 - The dashed fit curve updates after every click.
 - Use **Remove Last** to undo, **Clear Points** to start over.
- 4 Inspect the fit curve and RMS at the top right. The residual graph should look essentially random.
- 5 Click **Fit & Apply** - the correction is applied to the main dataset.

3.5 Recommendations

- Place points evenly: **both endpoints + center + any inflection regions**, typically 6 to 10 points.
- Clicking twice on the same wavelength replaces the previous click for that wavelength.
- Applying Background Correction first often makes the ridge easier to see.

§4. Kinetic Fit (single trace) (Slide 4)

4.1 Purpose

Fit a single-wavelength (or narrow-band-averaged) $\Delta A(t)$ trace with a **sum-of-exponentials + Gaussian IRF** model and extract lifetimes τ and amplitudes A. Stretched exponentials (β) are also supported.

4.2 Prerequisites

- A dataset must be loaded.
- (Recommended) Apply background, chirp, and any other needed corrections first.

4.3 Dialog layout

Left = Setup panel, right = trace + fit + residual graphs.

4.3.1 Setup

- **Centre λ (nm)** - center wavelength for the fit. **Use main λ** copies the crosshair λ from the main window.
- **$\pm$ half-width (nm)** - average over `centre  $\pm$  half-width` (0 = single pixel).
- **Components** - number of exponential components.
- **$\tau = \infty$ offset** - include a constant (infinite-lifetime) offset.
- **Components - initial values (ps)** table - per component:
- `$\tau_{init}$  (ps)` initial value

- τ fixed - hold τ constant during the fit
- Stretched - use stretched exponential; configures β_{init} and β fixed
- **IRF (Gaussian)**
- t_{IRF} (ps) plus its Fixed checkbox
- FWHM (ps) plus its Fixed checkbox
- Stretched-IRF mode - dropdown for how the IRF interacts with stretched exponentials (e.g. Skip (mask 3σ around t_{IRF})).
- **Fit window (delay)** - From / To. **Full** uses the entire delay range.
- **Run Fit** - run the optimization. **Reset** restores the initial values.
- **Export** - Trace+Fit (CSV) and Params (CSV).
- Bottom info: status line (Fit converged ... iters, RMS = ...), per-component τ / β / A / type, plus the estimated t_{IRF} and FWHM.

4.3.2 Graphs

- Top: data (black dots) and fit (red line).
- Bottom: residual (data - fit).

4.4 Procedure

- 1 Open **More Analysis** → **Kinetic Fit (single trace)**... on the toolbar.
- 2 Set the fit range via the **Fit window (delay)** From / To.
 - Too narrow misses long components; too wide may let baseline drift bias the fit.
- 3 Choose **Components** and enter approximate initial values for τ_{init} (e.g. 0.5, 3, 50 ps).
- 4 (Optional) Tick $\tau = \infty$ offset for a constant offset; turn on **Stretched** where applicable.
- 5 Click **Run Fit**.
- 6 Check the results
 - The red fit line should follow the data.
 - The **residual** panel should be roughly randomly distributed around zero (visible structure → model is too simple; add components or adjust the range).
 - Check that the reported RMS and τ values are physically reasonable.
- 7 If needed, export with **Trace+Fit (CSV)** and **Params (CSV)**.

4.5 Common problems

Symptom	Check / fix
Red fit clearly deviates from data	τ_{init} is far from reality → re-estimate by orders of magnitude (e.g. 1 ps instead of 100 ps).
Sinusoidal / step structure in residual	Too few components. Increase Components or enable $\tau = \infty$ offset.
t_{IRF} drifts strongly negative	Chirp correction missing - run §3 first.
Run Fit warns "n_min ..."	Number of points in fit window is too small for the chosen number of components → widen From / To.

Appendix A. Features NOT covered in this manual

This document is scoped to the 5 slides of the attached pptx. The following features exist in the program but require separate documentation (for reference):

- Crop Data, Mask Wavelengths
- Subtract solvent IRF
- Set $t = 0$ here
- Global Analysis
- SVD, LDA, MCR-ALS, Coherence (More Analysis menu)
- Load & Average (multi-file averaging)